The generative AI revolution has accelerated over the last few years, with advancements reshaping multiple industries by automating complex tasks and enhancing human capabilities. While proprietary AI models have long been the staple of tech giants, the rise of open source solutions has democratised the AI space, making powerful models like language models (LLMs), vision language models (VLMs), language action models (LAMs), speech-driven models (SLMs), and retrieval-augmented generation (RAG) agents accessible to all. Open source generative AI is breaking down barriers, offering unprecedented levels of transparency, customisation, and collaboration.

Understanding the key open source models in generative AI

Large language models (LLMs)

At the heart of generative AI lies the power of large language models (LLMs). These models, such as GPT (Generative Pretrained Transformer), are designed to understand and generate human language. LLMs are trained on a large corpora of text data, enabling them to answer questions, write essays, summarise documents, and even engage in sophisticated conversations. The open source movement has led to a proliferation of LLMs, enabling businesses, researchers, and developers to use, fine-tune, and scale these models to meet specific needs.

The key benefits of open source LLMs include:

Cost-effectiveness: Open source models like GPT-2, GPT-Neo, and GPT-J allow businesses to leverage advanced NLP capabilities without incurring hefty licensing fees.

Customisation: Open source models can be adapted and fine-tuned to specific domains, making them suitable for specialised use cases such as legal document generation, medical research, and customer service.

Visual language models (VLMs)

Visual language models (VLMs) combine natural language processing (NLP) with computer vision. These models are capable of understanding and generating both text and images, making them perfect for applications like caption generation, visual question answering (VQA), and image synthesis from textual descriptions. The open source community has made significant strides in developing models such as CLIP (Contrastive Language–Image Pretraining) and DALL-E that bridge the gap between vision and language.

Advantages of open source VLMs include:

Cross-modal understanding: Open source VLMs provide a framework for developing systems that can reason across both images and text, opening new doors for creative and analytical AI solutions.

Advanced content creation: Content creators are increasingly using these models for generating and modifying images based on textual input, which has profound implications for industries like marketing, e-commerce, and entertainment.

Language action models (LAMs)

Language action models (LAMs) are designed to understand not just language, but also the actions associated with it. These models can interpret natural language instructions and translate them into physical actions, making them ideal for applications in robotics, automation, and intelligent assistants. Open source LAMs, like the ones built on platforms such as OpenAI's Codex, allow for the creation of AI systems that can automate tasks across multiple domains by translating verbal commands into real-world actions.

The open source model's key benefits include:

Robotic Process Automation (RPA): With LAMs, robots or intelligent systems can learn from human instructions and execute complex tasks, ranging from industrial applications to home automation.

Interactive assistants: Open source LAMs power AI assistants capable of performing tasks such as scheduling meetings, controlling IoT devices, and even assisting in surgery.

Speech-driven models (SLMs)

Speech-driven models (SLMs) play a crucial role in converting speech to text and vice versa. These models have made significant contributions to fields like speech recognition, transcription, and voice-activated assistance. In the open source realm, projects like Mozilla's DeepSpeech and Kaldi have paved the way for highly accurate speech recognition and text-to-speech systems, driving the success of voice assistants like Siri, Alexa, and Google Assistant.

Key features of open source SLMs include:

Speech-to-text: These models can convert spoken language into written text with remarkable accuracy, transforming industries like healthcare, where transcribing medical records manually is time-consuming and prone to error.

Text-to-speech: Open source TTS models allow developers to create applications that can read out text aloud, useful in accessibility tools, e-learning platforms, and virtual assistants.

Retrieval-augmented generation (RAG) agents

Retrieval-augmented generation (RAG) agents are a powerful innovation in generative AI. They retrieve relevant information from large datasets or databases before generating a response, enhancing their accuracy and relevance. Open source RAG implementations,